

Design Project: Part 3

$$\text{Cutoff frequency} = 700 \text{ Hz}$$

$$\text{Minimum attenuation} = -15 \text{ dB at } 500 \text{ Hz}$$

$$\text{Passband gain} = 9 \text{ dB}$$

Butterworth-type response with maximally flat passband

① $\Rightarrow$ performing antilog calculation on attenuation:

$$10^{\frac{-15}{20}} = \underline{0.1778}$$

$$0.1778 = \frac{1}{\sqrt{1 + \left(\frac{f_c}{f}\right)^{2n}}}$$

$$(0.1778)^2 = \left(\frac{1}{\sqrt{1 + \left(\frac{700}{500}\right)^{2n}}} \right)^2$$

$$\left(1 + \left(\frac{7}{5}\right)^{2n}\right) \cdot 0.0316 = \frac{1}{1 + \left(\frac{7}{5}\right)^{2n}} \cdot \left(1 + \left(\frac{7}{5}\right)^{2n}\right)$$

$$\frac{\left(1 + \left(\frac{7}{5}\right)^{2n}\right) \cdot 0.0316}{0.0316} = \frac{1}{0.0316}$$

$$1 + \left(\frac{7}{5}\right)^{2n} = 31.6456$$

$$\left(\frac{7}{5}\right)^{2n} = 30.6456$$

$$\ln\left(\frac{7}{5}\right)^{2n} = \ln(30.6456)$$

$$\frac{2n \cdot \ln\left(\frac{7}{5}\right)}{2 \ln\left(\frac{7}{5}\right)} = \frac{\ln(30.6456)}{2 \ln\left(\frac{7}{5}\right)}$$

$$\boxed{n = 5.0858 \approx 6}$$

Figure 1: Task 1

② Using Sallen-Key

③ Converting passband gain:

$$10^{\frac{9}{20}} \approx 2.82$$

⇒ Adding a non-inverting op-amp to the end of the Sallen-Key will help realize the passband gain of 9 dB.

④

$$s = 0.9659 \pm j0.2588$$

Butterworth pole locations:

6th order: 0.9659 0.2588
 $s = -0.9659 \pm j(0.2588)$
 0.7071 0.7071
 $s = -0.7071 \pm j(0.7071)$
 0.2588 0.9659
 $s = -0.2588 \pm j(0.9659)$

$$(s + 0.9659 + j0.2588)(s + 0.9659 - j0.2588)$$

$$s^2 + 0.9659s - 0.2588js + 0.9659s + (0.9659)^2 + (0.9659 - 0.2588j) + 0.2588js + (0.2588j - 0.9659) - (0.2588j)^2$$

$$s^2 + 1.93s + 1 \quad \text{Assume: } C_1 = C_2 = 1 \text{ F}$$

$a=1$ $b=1.93$

$$b = R_1(C_1 + C_2) = 1.93$$

$$R_1(1+1) = 1.93$$

$$2R_1 = 1.93$$

$$R_1 = 0.965 \Omega$$

$$a = R_1 R_2 C_1 C_2 = 1$$

$$(0.965) R_2 (1)(1) = 1$$

$$R_2 = 1.04 \Omega$$

Impedance scaling: $K_m = 10k$

$$R_1' = (10k)(0.965)$$

$$R_2' = (10k)(1.04)$$

$$C_1' = \frac{1}{10k}$$

$$C_2' = \frac{1}{10k}$$

$$R_1' = 9.65 k\Omega$$

$$R_2' = 10.4 k\Omega$$

$$C_1' = 0.1 \mu F$$

$$C_2' = 0.1 \mu F$$

f scaling: $K_f = 2\pi(f_c) = 4400 \frac{\text{rad}}{\text{sec}}$

$$R_1'' = R_1' \rightarrow R_1'' = 9.65 k\Omega$$

$$C_1'' = \frac{0.1 \mu F}{4400}$$

$$R_2'' = R_2' \rightarrow R_2'' = 10.4 k\Omega$$

$$C_1'' = 0.0227 \mu F = C_2''$$

Figure 2: Tasks 2-4

$$s = 0.7071 \pm 0.7071j$$

$$(s + 0.7071 + 0.7071j)(s + 0.7071 - 0.7071j)$$

$$s^2 + 0.7071s - 0.7071js + 0.7071s + (0.7071)^2 - (0.7071)^2j + 0.7071js + (0.7071)^2j + (0.7071)^2$$

$$s^2 + 1.4142s + 1$$

$$a = 1 \quad b = 1.4142$$

Assume: $C_1 = C_2 = 1 \text{ F}$

$$b = R_1(C_1 + C_2) = 1.4142 \quad a = R_1 R_2 C_1 C_2 = 1$$

$$R_1(1+1) = 1.4142 \quad (0.7071)R_2(1)(1) = 1$$

$$2R_1 = 1.4142 \quad R_2 = 1.4142 \Omega$$

$$R_1 = 0.7071 \Omega$$

Impedance scaling: $K_m = 10k$

$$R'_1 = (10k)(0.7071) \quad R'_2 = (10k)(1.4142) \quad C'_1 = \frac{1}{10k} \quad C'_2 = \frac{1}{10k}$$

$$R'_1 = 7.07k\Omega \quad R'_2 = 14.14k \quad C'_1 = 0.1mF \quad C'_2 = 0.1mF$$

f scaling: $K_f = 4400 \frac{\text{rad}}{\text{sec}}$

$$R''_1 = R'_1 \rightarrow R''_1 = 7.07k \quad C''_1 = \frac{0.1mF}{4400}$$

$$R''_2 = R'_2 \rightarrow R''_2 = 14.14k \quad C''_1 = 0.02\mu F = C''_2$$

Figure 3: Task 4 Continued

$$s = -0.2588 \pm 0.9659j$$

$$(s + 0.2588 + 0.9659j)(s + 0.2588 - 0.9659j)$$

$$s^2 + 0.2588s - 0.9659js + 0.2588s + (0.2588)^2 - (0.9659j)(0.9659j) + 0.9659js + (0.2588)(0.9659j) + 0.9659$$

$$s^2 + 0.5176s + 1$$

$a=1$ $b=0.5176$ Assume: $C_1 = C_2 = 1 \text{ F}$

$$b = R_1(C_1 + C_2) = 0.5176 \quad a = R_1 R_2 C_1 C_2 = 1$$

$$R_1(1+1) = 0.5176 \quad (0.2588)R_2(1)(1) = 1$$

$$2R_1 = 0.5176 \quad R_2 = 3.86 \, \Omega$$

$$R_1 = 0.2588 \, \Omega$$

Impedance scaling: $k_m = 10k$

$$R'_1 = (10k)(0.2588) \quad R'_2 = (10k)(3.86) \quad C'_1 = \frac{1}{10k} \quad C'_2 = \frac{1}{10k}$$

$$R'_1 = 2.59 \text{ k}\Omega \quad R'_2 = 38.6 \text{ k}\Omega \quad C'_1 = 0.1 \text{ mF} \quad C'_2 = 0.1 \text{ mF}$$

f scaling: $k_f = 4400 \frac{\text{rad}}{\text{sec}}$

$$R''_1 = R'_1 \rightarrow R''_1 = 2.59 \text{ k}\Omega \quad C''_1 = \frac{0.1 \text{ mF}}{4400}$$

$$R''_2 = R'_2 \rightarrow R''_2 = 38.6 \text{ k}\Omega \quad C''_2 = 0.02 \mu\text{F} = C''_1$$

Figure 4: Task 4 Continued

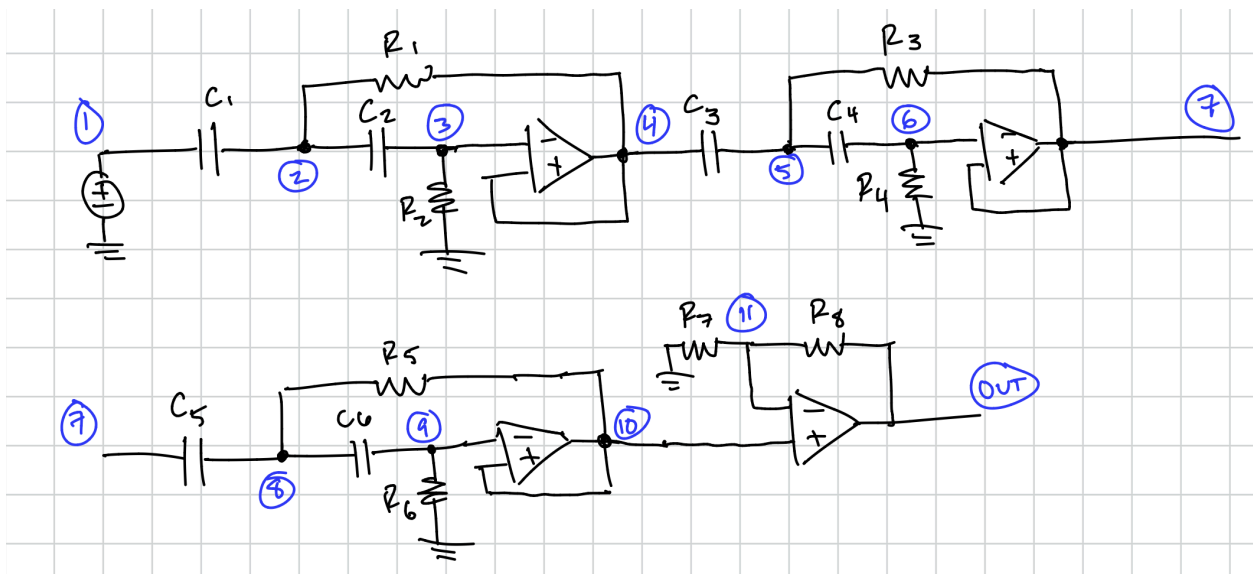


Figure 5: Complete Filter Circuit Schematic

```
Netlist 1

*Sallen-Key 1
V1 1 0 AC 1
C1 1 2 0.02u
C2 2 3 0.02u
R1 2 4 9.65k
R2 3 0 10.4k
X1 4 3 4 OPAMP

*Sallen-Key 2
C3 4 5 0.02u
C4 5 6 0.02u
R3 5 7 7.07k
R4 6 0 14.14k
X2 7 6 7 OPAMP

*Sallen-Key 3
C5 7 8 0.02u
C6 8 9 0.02u
R5 8 10 2.6k
R6 9 0 38.6k
X3 10 9 10 OPAMP

*Non-Inverting Op-Amp
R7 0 11 10k
R8 11 OUT 18.2k
X4 10 11 OUT OPAMP

.inc ideal_opamp.cir
.AC dec 500 100 100k
.end
```

Figure 6: Netlist #1

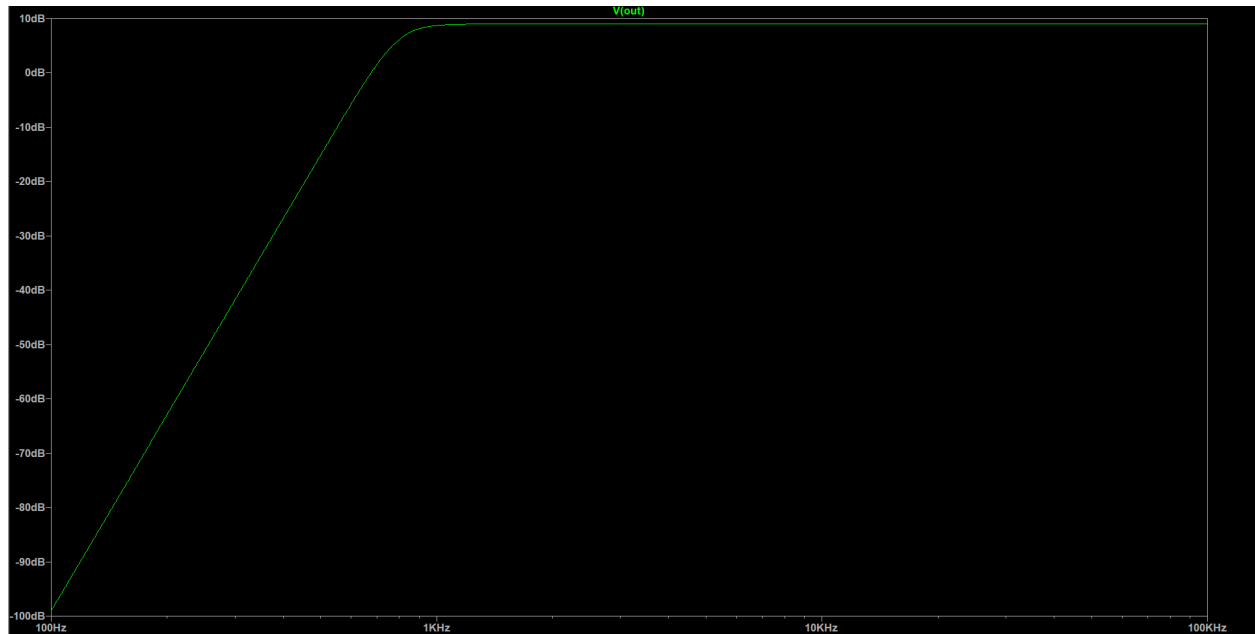


Figure 7: Bode Plot for Netlist #1

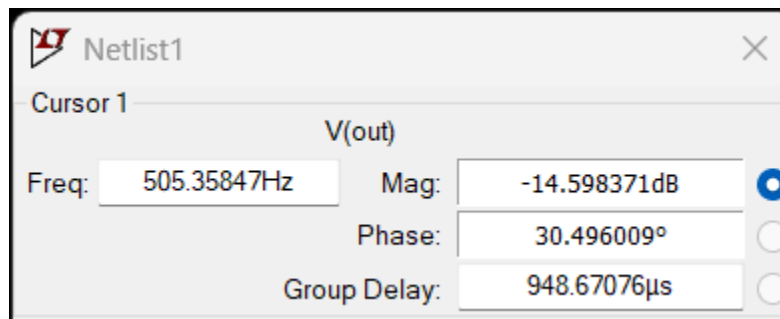


Figure 8: Attenuation at 500 Hz for Netlist #1

```

Netlist 2

.param tol_C1 = 0.2
.param tol_C2 = 0.2
.param tol_C3 = 0.2
.param tol_C4 = 0.2
.param tol_C5 = 0.2
.param tol_C6 = 0.2

*Sallen-Key 1
V1 1 0 AC 1
C1 1 2 {MC(0.02u, tol_C1)}
C2 2 3 {MC(0.02u, tol_C2)}
R1 2 4 9.65k
R2 3 0 10.4k
X1 4 3 4 OPAMP

*Sallen-Key 2
C3 4 5 {MC(0.02u, tol_C3)}
C4 5 6 {MC(0.02u, tol_C4)}
R3 5 7 7.07k
R4 6 0 14.14k
X2 7 6 7 OPAMP

*Sallen-Key 3
C5 7 8 {MC(0.02u, tol_C5)}
C6 8 9 {MC(0.02u, tol_C6)}
R5 8 10 2.6k
R6 9 0 38.6k
X3 10 9 10 OPAMP

*Non-Inverting Op-Amp
R7 0 11 10k
R8 11 OUT 18.2k
X4 10 11 OUT OPAMP
|
.inc ideal_opamp.cir
.step param run 1 500 1
.AC dec 500 100 100k
.end

```

Figure 9: Netlist #2

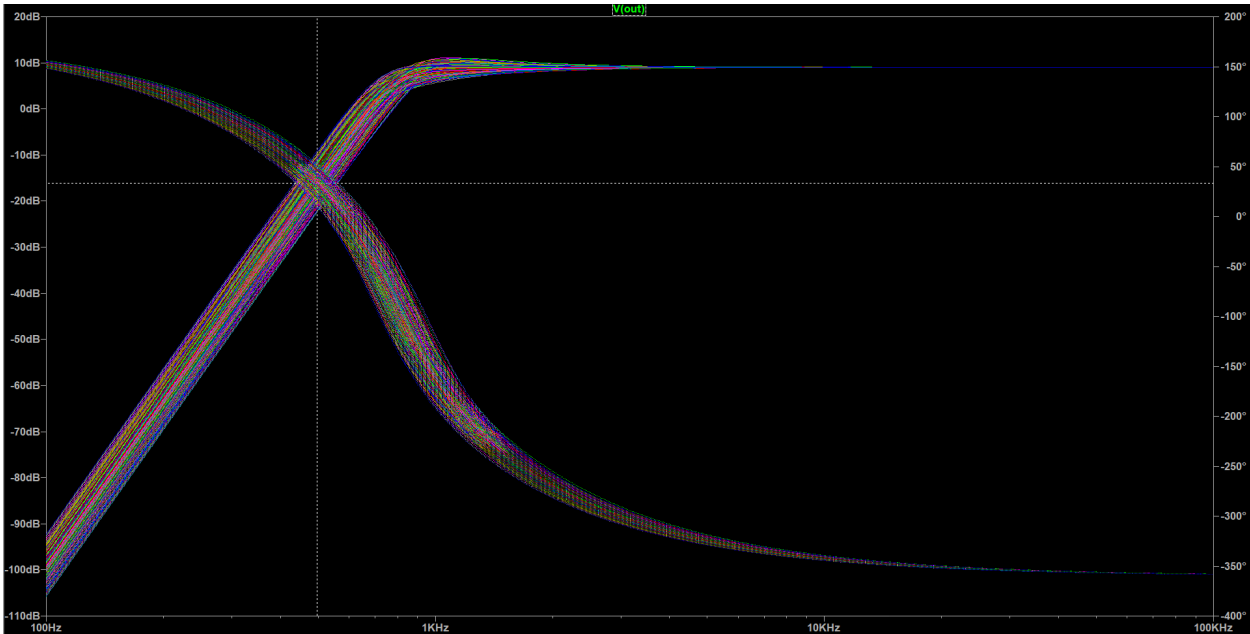


Figure 10: Bode Plot for Netlist #2

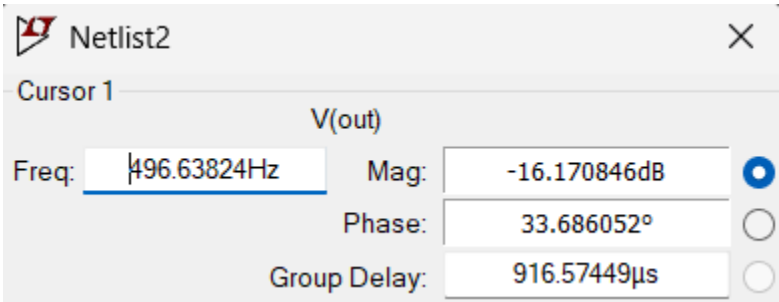


Figure 11: Attenuation at 500 Hz for Netlist #2


```

*EECE 311 Fall 2025 design project - template for filtered audio file output

*input voltage source for noisy birds recording: node 1 is positive node of voltage source
V1 1 0 wavefile=wind_birds.wav AC 1

*****

*Insert all nodes and elements for your filter circuit here

*Sallen-Key 1
C1 1 2 0.02u
C2 2 3 0.02u
R1 2 4 9.65k
R2 3 0 10.4k
X1 4 3 4 OPAMP

*Sallen-Key 2
C3 4 5 0.02u
C4 5 6 0.02u
R3 5 7 7.07k
R4 6 0 14.14k
X2 7 6 7 OPAMP

*Sallen-Key 3
C5 7 8 0.02u
C6 8 9 0.02u
R5 8 10 2.6k
R6 9 0 38.6k
X3 10 9 10 OPAMP

*Non-Inverting Op-Amp
R7 0 11 10k
R8 11 OUT 18.2k
X4 10 11 OUT OPAMP

*Don't forget the ".inc" commands for op-amp macromodels
.inc ideal_opamp.cir

*****

```

Figure 11: Netlist #3

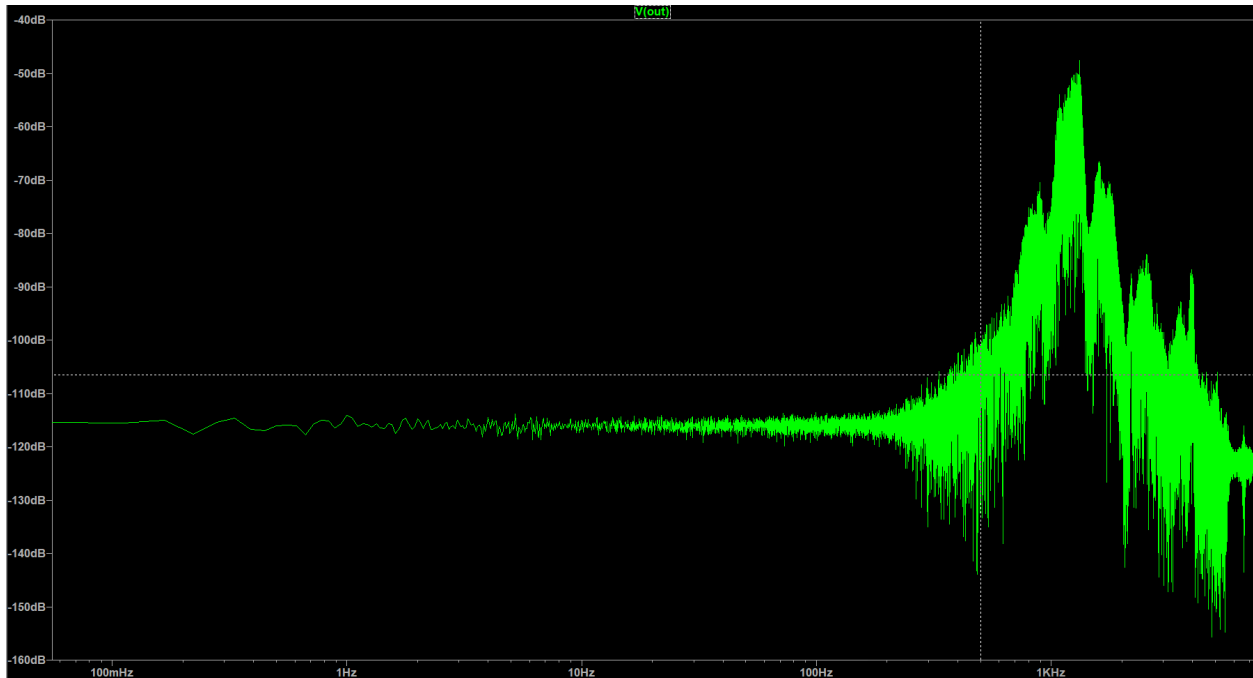


Figure 12: FFT Analysis for Netlist #3

9. Based on the MC Analysis that I performed in Task #6, there is a pretty considerable spread in the filter responses. Both the cutoff frequency and attenuation frequency had sizeable differences between the high end and the low end of the 20% tolerances. With that being said, I think it would make more sense to spend the extra money on capacitors with tighter tolerances, like 5% or 10%. When mass-produced, the likelihood of defective units increases dramatically, so I would want to try and mitigate the chances of that happening as much as possible.

```

clc; clear;

% Gain (9 dB)
K = 10^(9/20);

% Cutoff
fc = 700;
wc = 2*pi*fc; % rad/s

f = logspace(2,5,2000); % 100 Hz to 100 kHz
w = 2*pi*f; % rad/s

a1 = 0.5176;
a2 = 1.0000;
a3 = 1.9319;

num1 = [1 0 0];
den1 = [1 a1*wc wc^2];

num2 = [1 0 0];
den2 = [1 a2*wc wc^2];

num3 = [1 0 0];
den3 = [1 a3*wc wc^2];

num = K*conv(conv(num1,num2),num3);
den = conv(conv(den1,den2),den3);

H = freqs(num, den, w)

mag = 20*log10(abs(H));
ph = unwrap(angle(H))*180/pi;

% Magnitude plot
figure;
semilogx(f, mag, 'LineWidth', 1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('High-Pass Butterworth Filter (Magnitude)');

% Phase plot
figure;
semilogx(f, ph, 'LineWidth', 1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (deg)');
title('High-Pass Butterworth Filter (Phase)');

```

Figure 13: MATLAB Script

```

2.8172 + 0.0840i  2.8172 + 0.0837i  2.8173 + 0.0834i  2.8173 + 0.0831i  2.8173 + 0.0829i  2.8173 + 0.0826i
Columns 1945 through 1950
2.8173 + 0.0823i  2.8173 + 0.0820i  2.8173 + 0.0817i  2.8173 + 0.0814i  2.8173 + 0.0812i  2.8173 + 0.0809i
Columns 1951 through 1956
2.8173 + 0.0806i  2.8173 + 0.0803i  2.8173 + 0.0800i  2.8173 + 0.0798i  2.8174 + 0.0795i  2.8174 + 0.0792i
Columns 1957 through 1962
2.8174 + 0.0789i  2.8174 + 0.0787i  2.8174 + 0.0784i  2.8174 + 0.0781i  2.8174 + 0.0779i  2.8174 + 0.0776i
Columns 1963 through 1968
2.8174 + 0.0773i  2.8174 + 0.0771i  2.8174 + 0.0768i  2.8174 + 0.0765i  2.8174 + 0.0763i  2.8174 + 0.0760i
Columns 1969 through 1974
2.8175 + 0.0757i  2.8175 + 0.0755i  2.8175 + 0.0752i  2.8175 + 0.0750i  2.8175 + 0.0747i  2.8175 + 0.0744i
Columns 1975 through 1980
2.8175 + 0.0742i  2.8175 + 0.0739i  2.8175 + 0.0737i  2.8175 + 0.0734i  2.8175 + 0.0732i  2.8175 + 0.0729i
Columns 1981 through 1986
2.8175 + 0.0727i  2.8175 + 0.0724i  2.8175 + 0.0722i  2.8175 + 0.0719i  2.8175 + 0.0717i  2.8176 + 0.0714i
Columns 1987 through 1992
2.8176 + 0.0712i  2.8176 + 0.0709i  2.8176 + 0.0707i  2.8176 + 0.0704i  2.8176 + 0.0702i  2.8176 + 0.0700i
Columns 1993 through 1998
2.8176 + 0.0697i  2.8176 + 0.0695i  2.8176 + 0.0692i  2.8176 + 0.0690i  2.8176 + 0.0688i  2.8176 + 0.0685i
Columns 1999 through 2000
2.8176 + 0.0683i  2.8176 + 0.0680i
>>

```

Figure 14: Solved Transfer Function from MATLAB

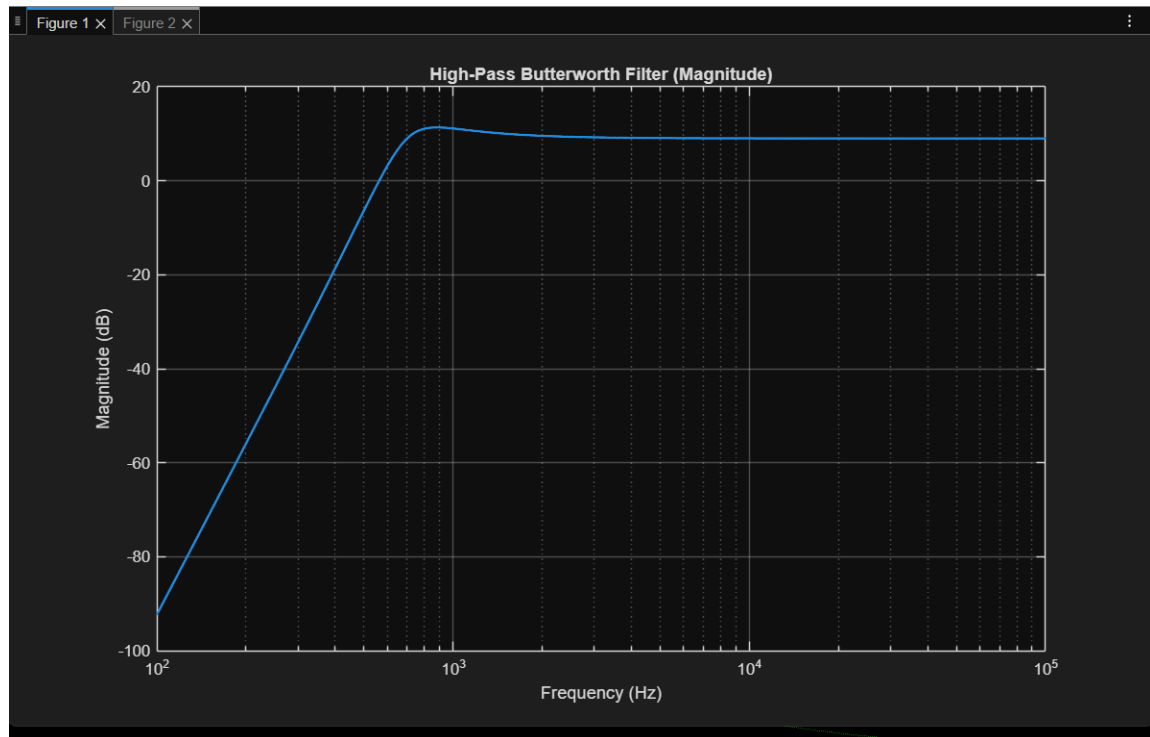


Figure 15: Magnitude Plot for Transfer Function

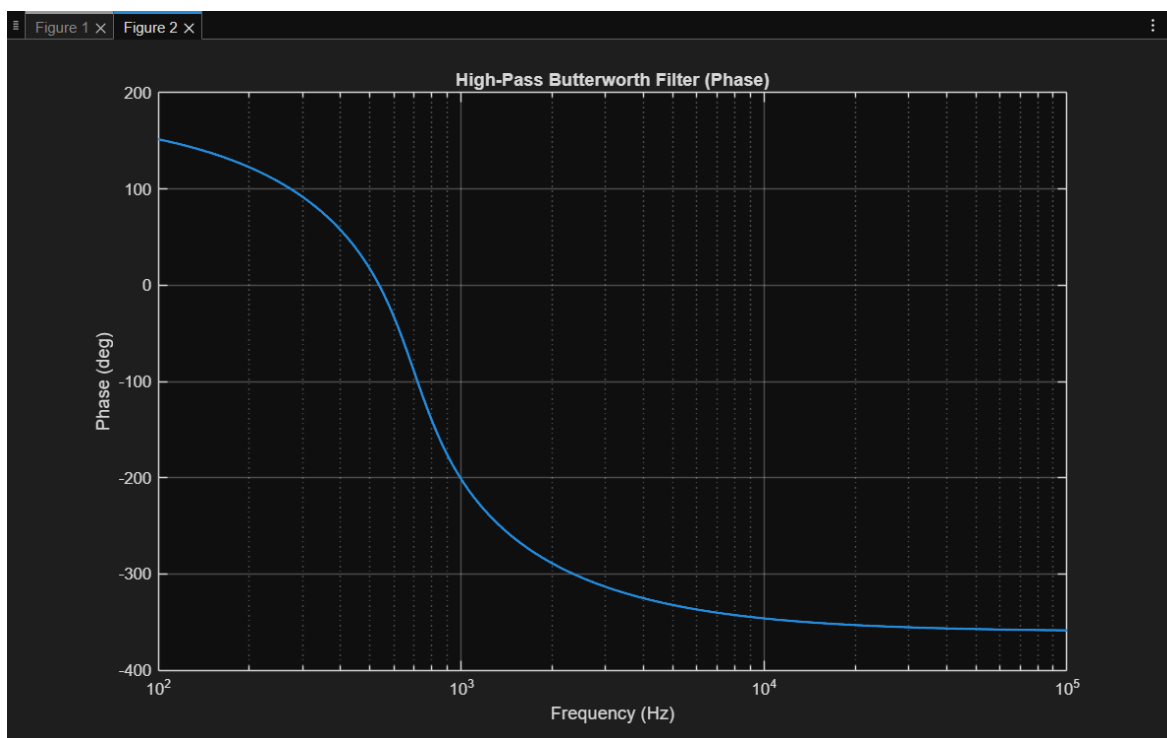


Figure 16: Phase Plot for Transfer Function